



Management Rules for Functional Testing Process 功能测试流程管理细则

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1.Introduction 引言

1.1 Purpose 目的

Based on the AWM Test Handbook, refine and clarify the functional testing process, achieve unity in the functional testing process within the organization, improve overall testing efficiency, and provide guidance for functional testing activities.

以 AWM Test Handbook 为基础，细化并明确功能测试流程，实现组织内功能测试过程的统一，提升整体测试效率，并为功能测试活动提供指导。

1.2 Definitions 术语定义

Term name 术语名称	Term description 术语描述
functional testing 功能测试	Based on the user needs, software features, and operational descriptions of the information system, test its characteristics and actionable behavior to determine that they meet user needs and design requirements. 根据信息系统的用户需求、软件特征、操作描述，测试其特性和可操作行为，以确定它们满足用户需求及设计要求。
Smoke testing 冒烟测试	It refers to testing the most basic functions of a new version after completing its development. 是指完成一个新版本的开发后，对该版本最基本的功能进行测试。
I-Test	Test refers to the testing conducted by testers on the functionality and interaction logic with other modules or systems in this release. I-Test 是指测试人员对本次发布的功能及与其他模块或系统的交互逻辑进行测试。
UAT	- User Acceptance Testing. 用户验收测试。 - Usually, it is a test conducted by the end software user to verify whether the system meets the actual business requirements based on actual business needs. 通常是由最终软件用户从实际业务出发，验证系统是否满足实际业务需求的测试。
E2E	- End To End testing. 端到端测试。 - Integrate multiple associated systems to achieve complete business process testing. 集合多个关联系统，实现完整的业务流程测试。



2. Main roles and responsibilities 主要角色及职责

Role name 角色名称	Responsibilities 职责描述
Product Owner 产品经理 (PDO/BA)	- Define product requirements, provide and explain product requirement manuals/user stories ; 定义产品需求，提供并讲解产品需求说明书/用户故事； - Review and confirm test cases ; 评审、确认测试用例； - Confirm test results ; 确认测试结果。
Agile Master 敏捷教练	- Promote the implementation of agile processes and methodologies to protect the team from interference during the sprint process; 推动敏捷过程和方法论的实施，保护团队在 sprint 过程中不受干扰； - Drive problem-solving 推动问题的解决； - Assist in promoting the improvement of process delivery quality ; 协助推动过程交付质量的提升。
Developers 开发人员	- Complete development self testing and provide test versions; 完成开发自测并提供测试版本； - Technical support for testing process; 测试过程技术支持； - Timely modification of defects during the testing process. 及时修改测试过程中的缺陷。
Operations personnel 运维人员 (L3-OPS)	- Build and maintain a system testing and operating environment; 搭建并维护系统测试、运行环境； - Release various environmental versions. 发布各环境版本。
Test Management 测试管理人员	- Interface person for testing requirements, responsible for receiving testing requirements and requirement changes; 测试需求接口人，负责接收测试需求和需求变更； - Participate in requirements review to understand the content and implementation methods of the requirements; 参与需求评审，了解需求内容和实现方式； - Clearly define the overall strategy for functional testing, including key business and important testing points, testing scenarios, etc; 明确功能测试总体策略，包括重点业务和重要测点、测试场景 - Develop a testing plan based on the project plan; 依据项目计划，制定测试计划； - Test admission review; 测试准入审核；



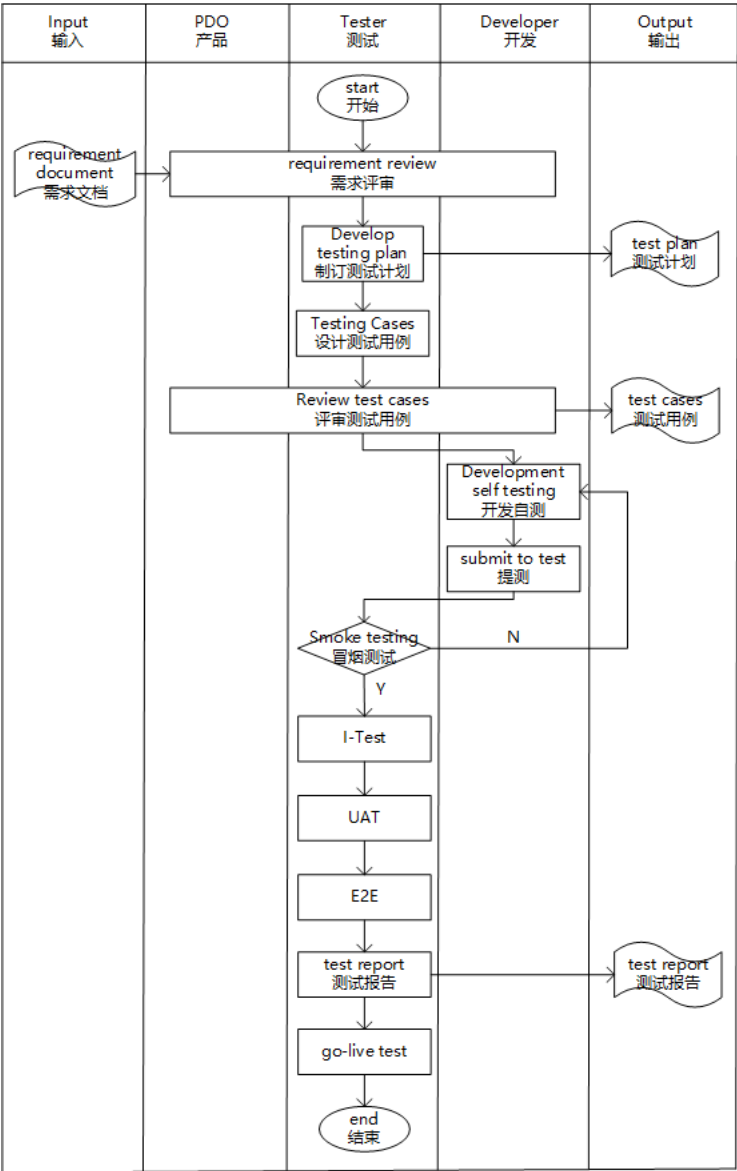
	- Review test cases; 评审测试用例； - Review the test execution results to determine whether they meet the exit criteria; 审核测试执行结果，判断是否符合准出标准； - Coordinate cross system testing and joint debugging matters; 协调跨系统测试联调事项； - Track the testing process, assist in solving problems during the testing process, and monitor risks during the testing process; 跟踪测试过程，协助解决测试过程中的问题，监控测试过程中的风险； - Track and manage test defects, output test reports. 跟踪和管理测试缺陷，输出测试报告。
Tester 测试执行人员	- Participate in requirement review and requirement explanation meetings ; 参加需求评审和需求讲解会议； - Analyze testing requirements, write test cases, and organize use case review meetings ; 分析测试需求，编写测试用例，组织用例评审会； - Perform specific testing tasks ; 执行具体测试任务； - Record, submit, and verify testing process defects, track defects until they are closed ; 记录、提交并验证测试过程缺陷，跟踪缺陷直至缺陷关闭； - Problems and risk warnings discovered during the testing process. 测试过程中发现的问题和风险预警。

3. Preconditions 前置条件

Entry Criteria 入口准则
- The requirements document review has been approved, and the approved requirements documents can be viewed on Jira 需求文档评审通过，Jira 上可查阅评审通过的需求文档 100% pass rate for developing self test cases 开发自测用例通过率 100%



4. Testing process 测试过程



4.1 Requirements Review and Analysis 需求评审与分析

Test management personnel and relevant testing personnel need to participate in the requirement refinement meeting and provide review and improvement suggestions for the requirement document. 测试管理人员与相关测试人员需参加需求 refine meeting，对需求文档提出评审和改进意见。

4.2 Develop a testing plan 制订测试计划

Test management personnel organize testers to analyze testing requirements, develop testing plans and plans, determine testing scope (including functional analysis and correlation analysis), testing



environment, and testing data, and communicate with the development team and PDO to determine testing plans and plans. The testing plan and plan should specify: testing scope, testing plan cycle, environment/time/content/method of joint debugging with other systems, testing personnel ratio, testing workload, testing risks, etc.

测试管理人员组织测试人员分析测试需求，制定测试计划与方案，确定测试范围（包括功能分析、关联性分析）、测试环境及测试数据，并与开发团队、PDO 沟通确定测试计划与方案。测试计划与方案中需明确：测试范围、测试计划周期、与其他系统联调的环境/时间/内容/方式、测试人员配比、测试工作量、测试风险等。

After the development of the test plan is completed and reviewed, the results of the 《Test Plan》 will be output.

测试计划制订完成并通过评审后，输出《测试计划》成果物。

4.2.1 Determine testing strategy 确定测试策略

The testing strategy generally describes the general methods and objectives of software testing activities. Including the testing steps to be conducted (I-Test, UAT testing, etc.) and the types of tests to be executed (functional testing, performance testing, security testing, automation testing, etc.)

测试策略一般描述软件测试活动的一般方法和目标。包括要进行的测试环节（I-Test 测试、UAT 测试等）以及要执行的测试类型（功能测试、性能测试、安全测试、自动化测试等）。

4.2.2 Estimated testing workload 预估测试工作量

Testers estimate the testing workload based on specific requirements, testing scope, and testing strategies. The testing workload should cover requirements understanding, test case design, test data preparation, and various testing activities execution.

测试人员结合具体的需求、测试范围及测试策略，预估测试工作量，测试工作量应覆盖需求理解及测试用例设计、测试数据准备及各类测试活动执行等。

4.2.3 Test Timeline 测试时间线

Refer to the overall project timeline and estimate the testing workload to output timelines for various testing activities. Please refer to the 《Testing Plan》

参考项目整体 timeline，结合预估测试工作量，输出各类测试活动的 timeline，可参考《测试计划》。

4.2.4 Estimate testing risks 预估测试风险

During the testing execution process, there may be various situations that affect the testing progress, namely risks

测试执行过程中，可能会出现各类影响测试进度的情况，即风险。

In the testing plan, predict these risks in advance and provide corresponding measures. Common risks and solutions are as follows:

测试计划中应对这些风险提前做出预估并给出应对措施，常见的风险及解决方案如下：



1) Smoke testing failed

冒烟测试未通过

- Risk analysis: If the test version submitted by the development team fails the smoke test, it will result in resubmitting the test and delaying the testing progress.
风险分析：如果开发提交的测试版本，执行冒烟测试未通过，则会导致再次提交测试，造成测试进度延期。
- Solution: Before submitting the test version, developers should conduct self testing to ensure 100% pass rate of self testing test cases before submitting them to testers.
解决方案：在提交测试版本前，开发应进行自测，确保自测测试用例通过率 100%后再提给测试人员。

2) Delayed submission of testing during development

开发提测延误

- Risk analysis: Delayed development and testing lead to delayed or compressed testing work, resulting in the risk of overall testing work completion being delayed and insufficient testing.
风险分析：开发提测延期导致后续测试工作顺延或压缩测试时间，从而造成整体测试工作完成延期及测试不充分的风险。
- Solution: Developers are requested to strictly control the progress during the project. If there is a risk of delay, please notify the testers immediately and negotiate a solution. If the testing cannot be conducted on time after the expiration date, the testing management personnel need to evaluate whether they can absorb the risk of delayed testing in the development process. If necessary, they should apply to the project team to modify the testing plan and synchronize it with relevant personnel.
解决方案：请开发人员在项目进行过程中严格控制进度，如果有推迟的风险请立即通知测试人员，协商解决。如到期仍然不能按时提测，则测试管理人员需要评估能否消化开发延期提测的风险，必要时向项目组申请修改测试计划，并同步给相关人员。

3) Critical or High defect blocking test

严重或高级别缺陷阻塞测试

- Risk analysis: During the test execution process, if critical or high defects are found in the tested version that hinder the test execution, if they are not fixed in a timely manner, the test schedule will be affected.
风险分析：测试执行过程中，被测试版本发现 Critical 或 High 且阻碍测试执行的 defect 时,如果修复不及时，测试进度安排受到影响。
- Solution: Develop priority solutions for critical or high defects, adjust test case execution strategies by test management personnel, and upgrade block issues to the project team.
解决方案：开发优先解决 Critical 或 High 缺陷，测试管理人员调整测试用例执行策略，同时将 block 问题升级到项目组。

4.3 test case design 测试用例设计

Test execution personnel write test cases based on product requirements, which need to include elements such as test function points, test conditions, test steps, input values, expected results,



priority, severity, etc. Test cases need to cover all scenarios of the requirements and consider special scenarios such as various anomalies and boundary conditions.

测试执行人员依据产品需求编写测试用例，测试用例需要包含测试功能点、测试条件、测试步骤、输入值、预期结果、优先级、严重程度等要素。测试用例需覆盖需求的全部场景，并考虑各种异常情况、边界情况等特殊场景。

Please refer to the 《Test Case Design Guide》 for details
详见《测试用例设计指南》。

4.4 Test Case Review 测试用例评审

Test management personnel organize test case reviews, and both developers and product managers need to participate in the review. The review of test cases is conducted through formal meetings or offline review through email countersigning. The use case review meeting needs to clarify the scope of this requirement testing and smoke test cases, whether there are any functional omissions, whether the description is clear, whether the test data is accurate, and reach an agreement with relevant stakeholders. After the review of the test cases is completed, the testers will modify the test cases based on the review suggestions. After updating, they will send an email to the relevant parties and clarify the smoking test cases required for this time. It is recommended to develop a self-test based on this.

测试管理人员组织测试用例评审，开发人员和产品经理均需参与评审。测试用例评审通过正式会议方式进行或邮件会签的离线评审方式进行。用例评审会需明确本次需求测试范围及冒烟测试用例范围，是否有功能遗漏、是否描述清晰、测试数据是否恰当，并与相关干系人达成一致。测试用例评审完成后，测试人员针对评审建议修改测试用例，更新后发送邮件给相关方并明确本次需求的冒烟测试用例，建议开发据此自测。

4.5 Test Start Check 测试启动检查

The test management personnel are responsible for conducting test initiation checks to determine whether the test execution work can start normally. The test initiation check includes checking whether the test environment and data are available. The test management personnel shall provide risk warnings (such as risk warnings for test delays or changes in test scope) for any abnormal situations detected before starting the test, and provide feedback to relevant stakeholders.

测试管理人员负责进行测试启动检查，确定测试执行工作是否可以正常启动。测试启动检查包括测试环境与数据是否具备等。测试管理人员对启动测试前检查的异常情况，进行风险提示（如测试延期或测试范围变更等风险提示），并反馈给相关干系人。

4.6 test execution 测试执行

4.6.1 Smoke Testing 冒烟测试

Smoke testing refers to the rapid verification of the basic functions, main scenarios, and main processes of the test version by testers to confirm the quality of the test version. On the one hand, it can avoid blocking the test process due to poor quality of the test version, and on the other hand, it



can reduce the repair and verification costs caused by excessive basic defects flowing into the testing phase.

冒烟测试是指由测试人员对提测版本的基本功能、主要场景和主要流程进行快速的验证，用于确认提测版本的质量，一方面，可以避免因提测版本质量过差而导致测试流程 block，另一方面，可以减少过多的基础 defect 流入到测试阶段而带来的修复成本和验证成本。

4.6.2 I-Test

I-Test is a functional test conducted by testers on the newly added or modified functions in a testing environment. The testing scope mainly includes logical verification of the newly added or modified functions, UI testing, and checking whether the interaction logic with other modules or systems is correct.

I-Test 是由测试人员在测试环境对本次新增或修改的功能进行功能测试，测试范围主要包括新增或修改功能的逻辑验证，UI 测试以及与其他模块或系统的交互逻辑是否正确。

4.6.3 UAT

It is a type of testing performed by key user for verification and acceptance testing before moving a software system to a production environment, conducted in a UAT environment to confirm whether business objectives have been achieved and user requirements have been met.

是一种由 KU 执行的测试，用于在将软件系统移至生产环境之前的验证、接收测试，在 UAT 环境进行，以确认是否实现了业务目标、满足用户需求。

UAT is carried out after the completion of I-Test, and test cases are prepared by testers or KU and jointly reviewed (Development&Testing&PDO&KU). For the preparation of test accounts and test data, SAP projects are generally prepared by consultants, while non SAP projects are prepared by testers. If necessary, developers provide technical support.

UAT 在 I-Test 测试完成后进行，由测试人员或 KU 准备测试用例且通过共同评审（开发&测试&PDO&KU），对于准备测试账号和测试数据的准备，一般情况下 SAP 项目由顾问准备，non-SAP 项目由 tester 准备，必要时开发人员给予技术支持。

During the UAT testing process, after the business customer raises a question, it is necessary for the testing side to confirm whether it is a defect or a requirement change. If it is a defect, it is maintained in the jira system and modified by the development personnel. After being verified by the testing personnel, it is submitted to the customer for regression verification. If it is a requirement change (confirmed by PDO), development and testing personnel need to evaluate the workload and reschedule.

在 UAT 测试过程中，业务客户提出问题后，需要由测试侧确认是缺陷还是需求变更，如果是缺陷，维护到 jira 系统，由开发人员修改，测试人员验证通过后提交客户进行回归验证。如果是需求变更（经 PDO 确认），需要开发和测试人员评估工作量后重新排期。

4.6.4 E2E

E2E is conducted in the UAT environment after UAT. The E2E test is triggered by the project team, with a focus on the actual business process. The integration of this launch function with all its



associated and dependent systems ensures the correctness of the business process and that the original functions are not affected by incremental changes.

E2E 是在 UAT 之后，在 UAT 环境进行。E2E 测试由项目组触发，测试重点为实际的业务流程，将本次上线功能与其所有关联、依赖的系统进行整合，确保业务流程的正确性及原有功能不受增量的影响。

4.6.5 Go-live test

Go-live test is an action taken by testers or KU to verify the online functionality after the system is launched. If problems are found during the verification process, the development team needs to make corresponding modifications in a timely manner. Test management personnel should organize and coordinate the testing process and follow up on the verification results.

Go-live test 为系统上线后,由测试人员或 KU 对上线功能进行验证的动作，如果验证过程中发现问题，开发需及时对应修改，测试管理人员要组织与协调测试过程、跟进验证结果。

4.6.6 brief summary 小结

For the testing implementation of a new project, the optional testing process is as follows:

对于一个新项目的测试实施工作，可选择的测试流程如下：

- Joint debugging and testing work for non rollout projects involving multiple systems. The recommended testing process is: development self testing→smoke testing→I-Test→UAT→E2E→Go-live test.
非 rollout 项目且涉及多个系统的联调测试工作，建议的测试流程：开发自测→冒烟测试→I-Test→UAT→E2E→Go-live test。
- Non rollout projects do not involve joint debugging and testing between systems. The recommended testing process is: development self testing → smoke testing → I-Test → UAT → Go live test.
非 rollout 项目不涉及系统间的联调测试工作，建议的测试流程：开发自测→冒烟测试→I-Test→UAT→Go-live test。
- The rollout project involves joint debugging and testing of multiple systems. The recommended testing process is I-Test → UAT → E2E → Go live test.
Rollout 项目且涉及多个系统的联调测试工作，建议的测试流程：I-Test→UAT→E2E→Go-live test。
- The rollout project does not involve system to system joint debugging and testing work. The recommended testing process is I-Test → UAT → Go live test.
Rollout 项目且不涉及系统间的联调测试工作，建议的测试流程：I-Test→UAT→Go-live test。

4.7 test summary 测试总结

During the testing process, it is recommended to send a Test Report every day to synchronize the testing progress, testing quality, and risk warning with the project team. The specific content can be found in the 《Test Report》



在测试过程中，建议每天发送 Test Report,用于向项目组同步测试进度、测试质量及风险预警，具体的内容见《Test Report》。

After the iteration is completed or the project is launched, it is recommended to conduct a review and summary, which can include communication and collaboration, various processes, requirements&development&testing quality, etc.

在迭代结束或项目上线后，建议进行复盘总结，复盘内容可包括沟通与协作，各类流程，需求&开发&测试质量等。

4.8 Post launch support 上线后支持

After entering hypercare, if any production defects are found, testing personnel need to verify the defects. After confirming the defects, the development team will fix them, and after passing the testing, the hotfix launch process will begin. At the same time, problems that arise during the Hypercare stage will be collected, sorted, classified (non error, improper operation, data issues, defects, etc.), and RootCause analysis will be conducted to propose corresponding strategies to improve the testing process and improve testing quality.

在进入 hypercare 后，如果发现生产 defect，需要测试人员对 defect 进行验证，确认 defect 后，由开发修复，测试通过后进入 hotfix 上线流程，同时对 hypercare 阶段出现的问题进行收集、整理、分类（非错、操作不当、数据问题、缺陷等等）并进行 RootCause 分析，提出应对策略，以此来改善测试过程、提升测试质量。

5. Defects Definition 缺陷定义

The so-called software defect refers to a problem, error, or hidden functional defect that disrupts the normal operation ability of computer software or programs. The existence of defects can lead to software products being unable to meet the needs of users to some extent. From an internal perspective, defects refer to various problems such as errors and defects that exist in the development or maintenance process of software products; From an external perspective of the product, a defect is a failure or violation of a certain function that the system needs to implement.

所谓软件缺陷，即为计算机软件或程序中存在的某种破坏正常运行能力的问题、错误，或者隐藏的功能缺陷。缺陷的存在会导致软件产品在某种程度上不能满足用户的需要。从产品内部看，缺陷是软件产品开发或维护过程中存在的错误、毛病等各种问题；从产品外部看，缺陷是系统所需要实现的某种功能的失效或违背。

5.1 Defect Severity 缺陷严重程度

The severity of a defect refers to the degree to which the failure caused by the defect affects the software product.

缺陷严重程度是指因缺陷引起的故障对软件产品的影响程度。

- Critical
Impact on production, or defects that make the system almost unusable or testing unable to continue, requiring immediate repair.



影响生产，或缺陷导致系统几乎不能使用或测试不能继续，需立即修复。

- **High**
The serious impact of defects on testing needs to be given priority consideration.
缺陷严重影响测试需要优先考虑。
- **Medium**
The defect is normally queued for repair.
缺陷正常排队等待修复。
- **Low**
Defects can be corrected within the developer's time.
缺陷可以在开发人员时间容许的情况下修订。

5.2 priority 优先级

The priority of defects refers to the urgency with which they must be repaired.

缺陷的优先级指缺陷必须被修复的紧急程度。

- **Critical**
The program cannot work and the test cannot be executed.
程序无法工作，测试无法执行。
- **High**
The main function has serious defects, but it will not affect system stability.
主要功能存在严重缺陷，但不会影响到系统稳定性。
- **Medium**
Does not affect the basic use of the system, but it has not been well implemented and did not achieve the expected results.
不影响系统的基本使用，但没有很好的实现功能，没有达到预期的效果。
- **Low**
Usability and advisory issues.
易用性及建议性问题。

5.3 Defect status 缺陷状态

State 状态	illustrate 说明
New	• Create an initial state for the issue, assign it to the developer, and wait for processing 创建问题初始状态，指派给开发，等待处理
In progress	• Developing and handling issues 开发处理问题中
In test	• Developed and modified issues, tested and verified 开发修改完的问题，测试验证
Resolved	• The development has modified the issue and submitted the version



	开发已修改问题并提交了版本
Closed	Confirm the repaired defects and close them 确认被修复的缺陷，将其关闭

5.4 Introduction phase 引入阶段

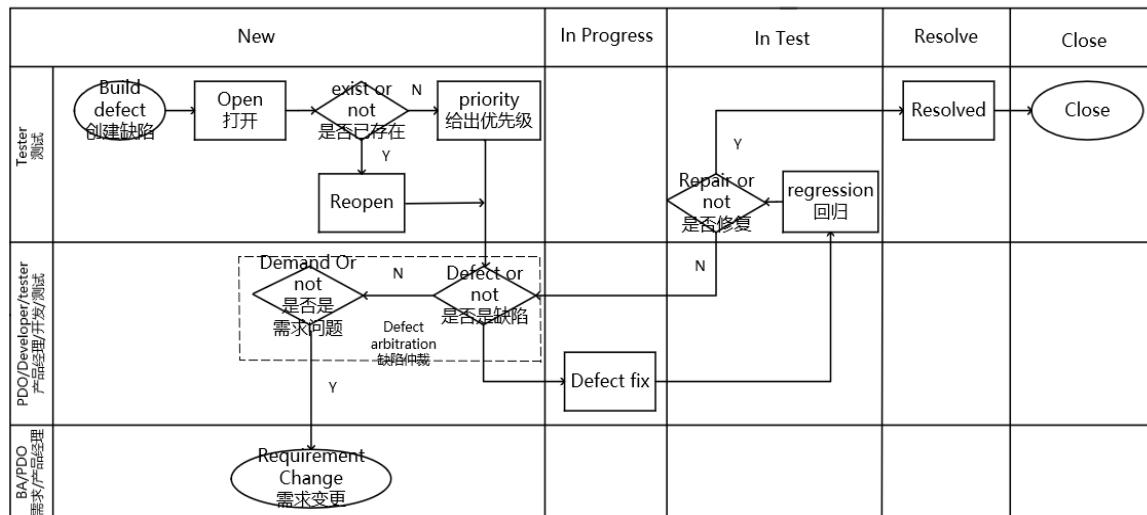
Introduction phase 引入阶段	illustrate 说明
I-Test	Test phase I-Test 阶段
UAT	UAT stage UAT 阶段
E2E	E2E stage E2E 阶段
Release	After going online but not delivered to the pre operation and maintenance hypercare stage 上线后未交运维前 hypercare 阶段
prod	Prod is the stage after transportation, operation and maintenance 交运维之后的阶段

5.5 Resolution time 解决时效

Resolution time 解决时效	
Critical	24h
High	24h
Medium	48h
Low	72h

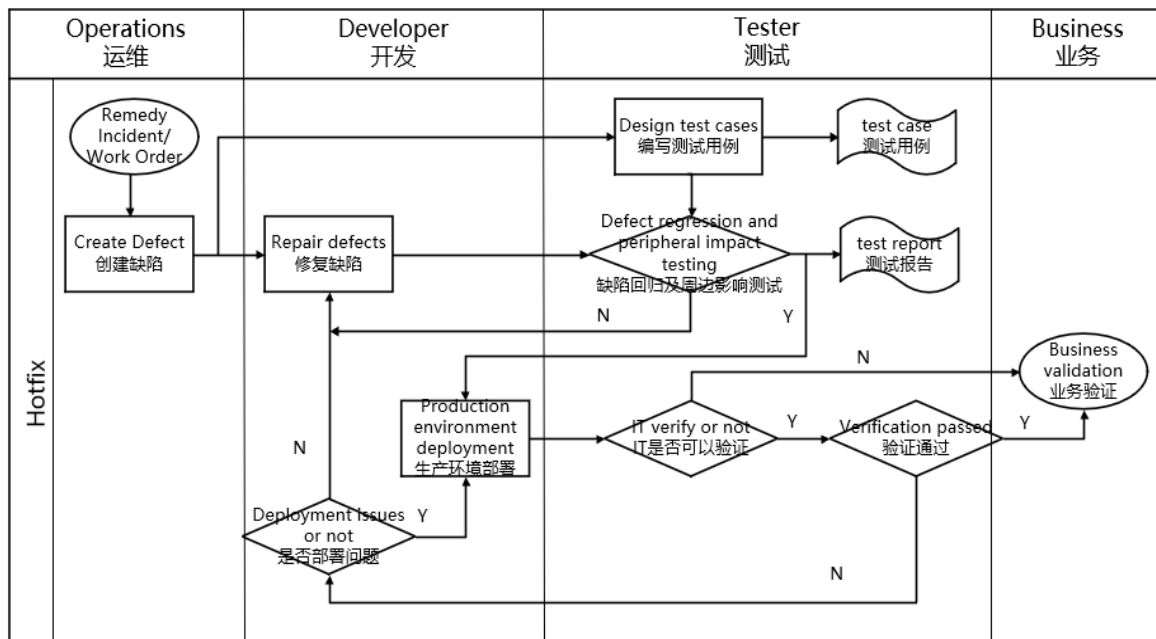
5.6 Defect circulation 缺陷流转

The defect flow process during the regular iteration process is as follows:
常规迭代过程中的缺陷流转流程如下：



The defect flow process for Hotfix is as follows:

对于 Hotfix 的缺陷流转流程如下：



6. REAR OUT 后置输出

6.1 Suspension criteria 中止准则

Discontinue 中止
Smoke test failed (pass rate not reaching 100%) 冒烟测试未通过（通过率未达到 100%）



- Unclear or inconsistent requirements
需求不明确或未达成一致

6.2 Exit criteria 出口准则

Exit criteria 出口准则
The coverage of test cases to requirements reaches 100% 测试用例对需求的覆盖率达到 100%
Test case execution rate reaches 100% 测试用例执行率达到 100%
Test case pass rate $\geq 95\%$ 测试用例通过率 $\geq 95\%$
All discovered defects are recorded and tracked 所有发现的缺陷均被记录并跟踪
Defect repair rate $\geq 95\%$ 缺陷修复率 $\geq 95\%$
- There are no critical and high level defects left behind 没有遗留 Critical 和 High 级别的缺陷 - Remaining Medium & Low defects $\leq 5\%$ 遗留的 Medium & Low 缺陷 $\leq 5\%$

6.3 output 输出

Output 输出
《Test Plan》 《测试计划》
《test case》 《测试用例》
《Test Case Review Record》 《测试用例评审记录》
《Functional Test Report》 《功能测试报告》